DLX Microprocessor - Design and Development

Microelectronic Systems Final Project

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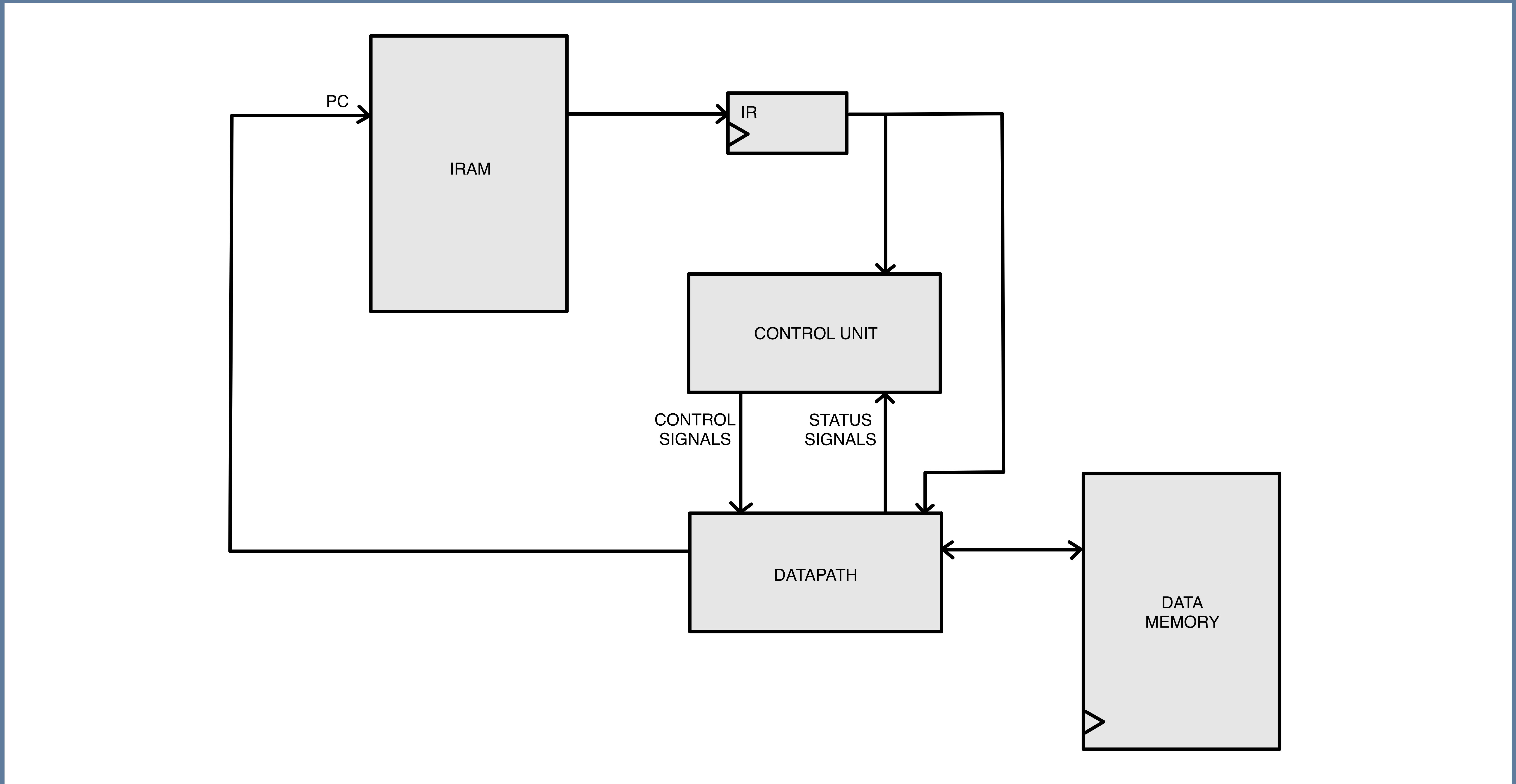
DLX Architecture

Overview

- RISC architecture (Hennessy & Patterson)
- 32-bit registers
- 5-stages pipeline
- 32 general purpose registers

DLX Architecture

Block Diagram



Instruction Set Architecture

- R-TYPE: *add, sub, and, or, xor, sll, srl, sne, sge, sle*
- I-TYPE: *addi, subi, andi, ori, xori, slli, srli, snei, sgei, slei, lw, sw, beqz, bnez*
- J-TYPE: *j, jal*

- Fixed 32-bit encoding
- Specific OPCODE assigned to each instruction

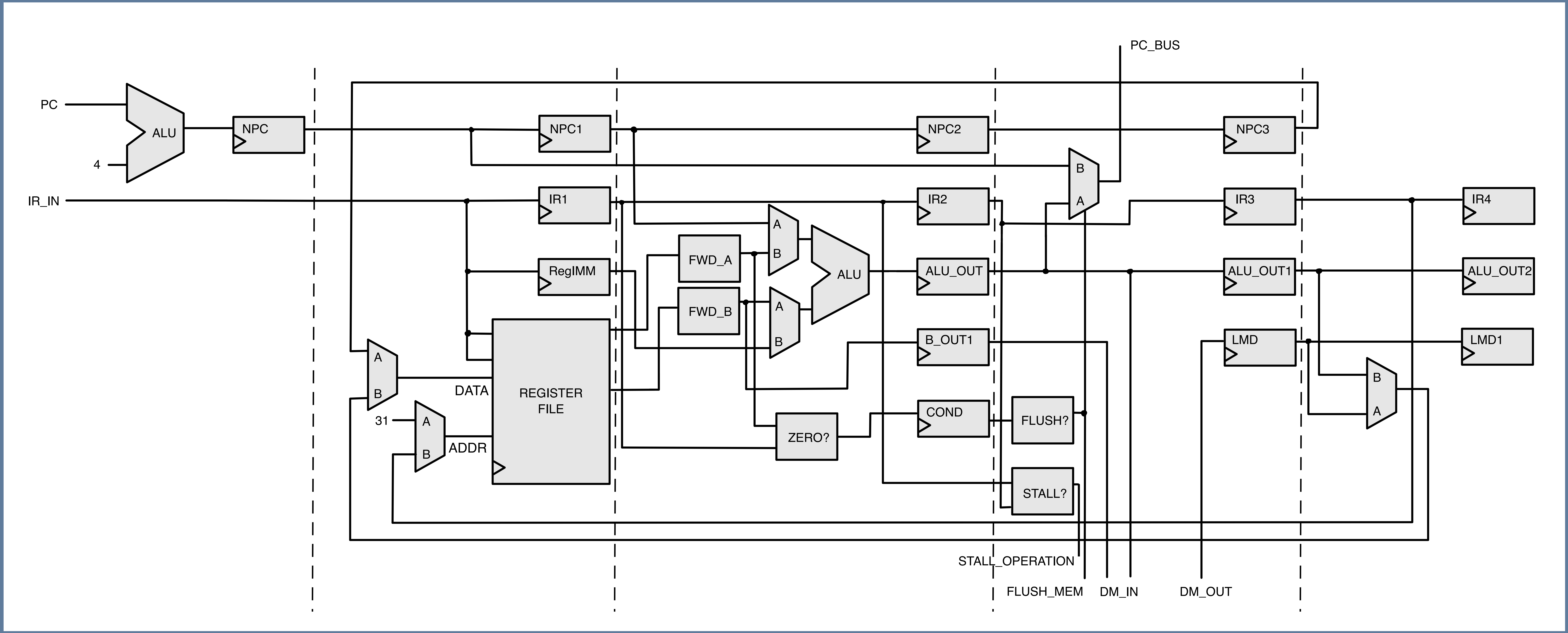
Datapath

Pipeline stages

- Instruction Fetch (IF)
- Instruction Decode (ID)
- Execute (EX)
- Memory (MEM)
- Write Back (WB)

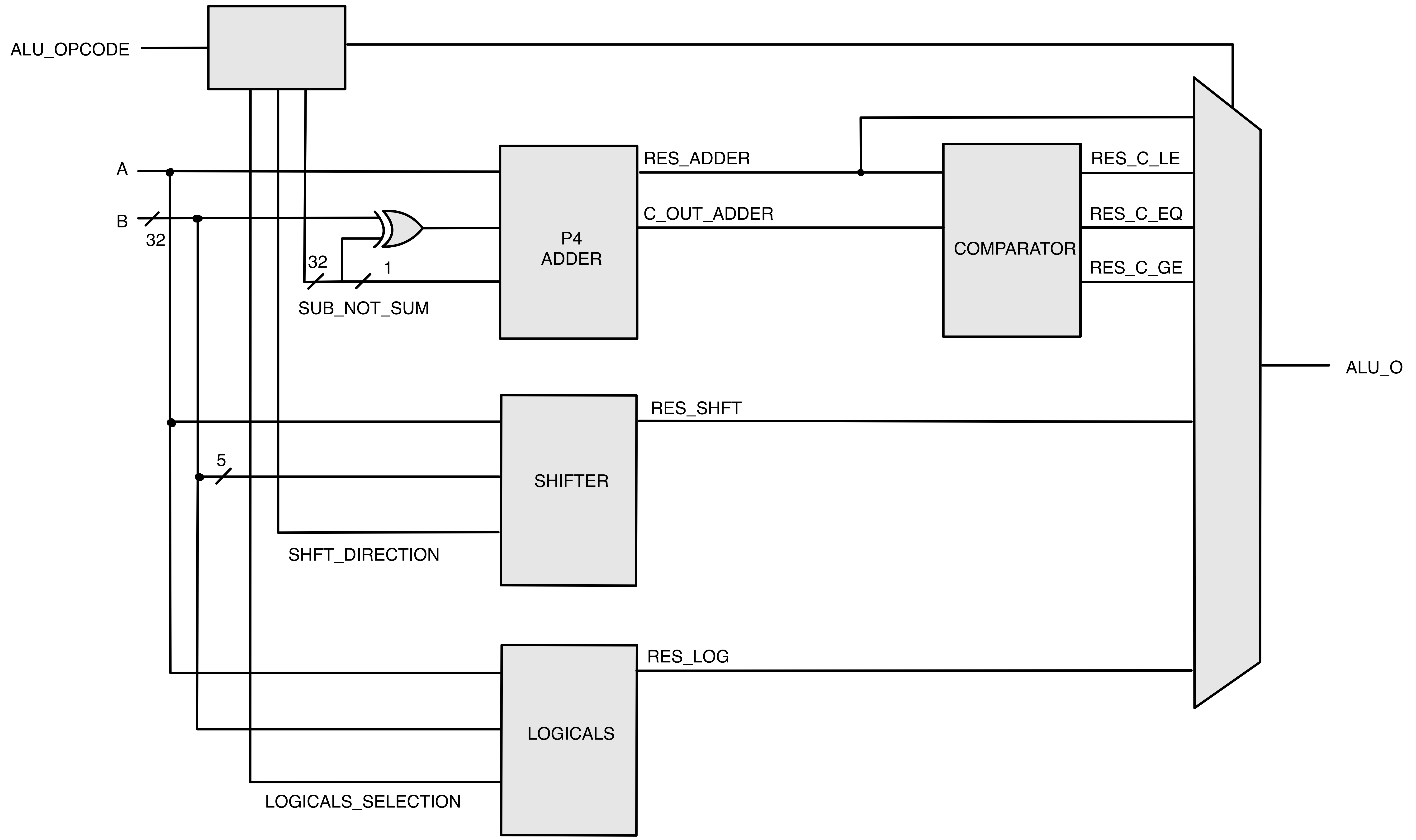
Datapath

Block Diagram



Datapath

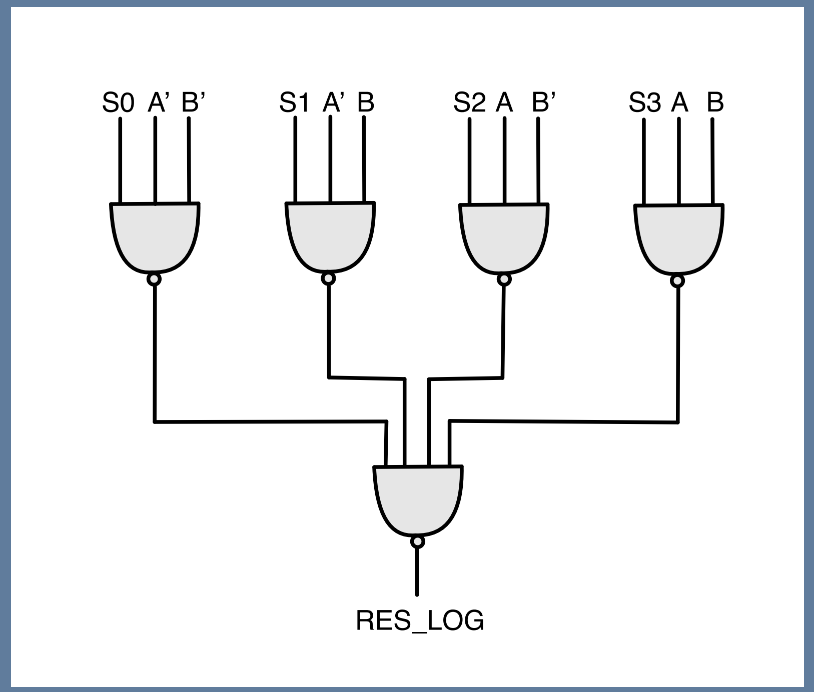
ALU - Block Diagram



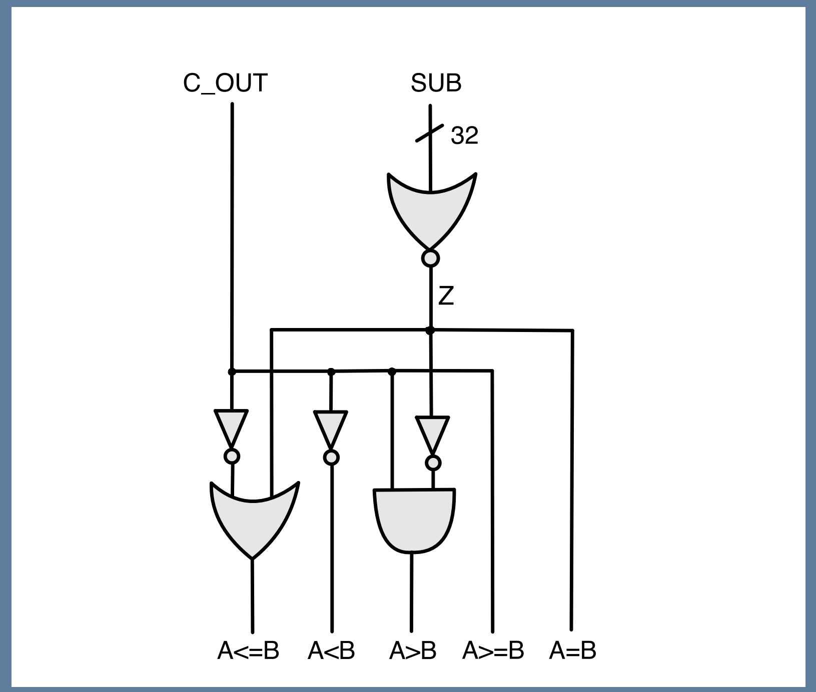
Datapath

ALU - Sub-blocks

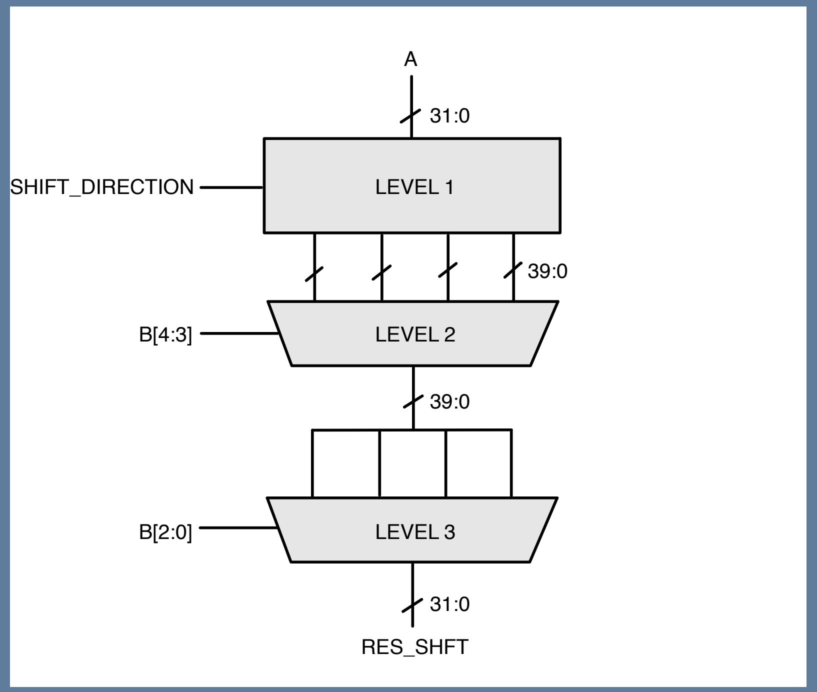
Logical Unit



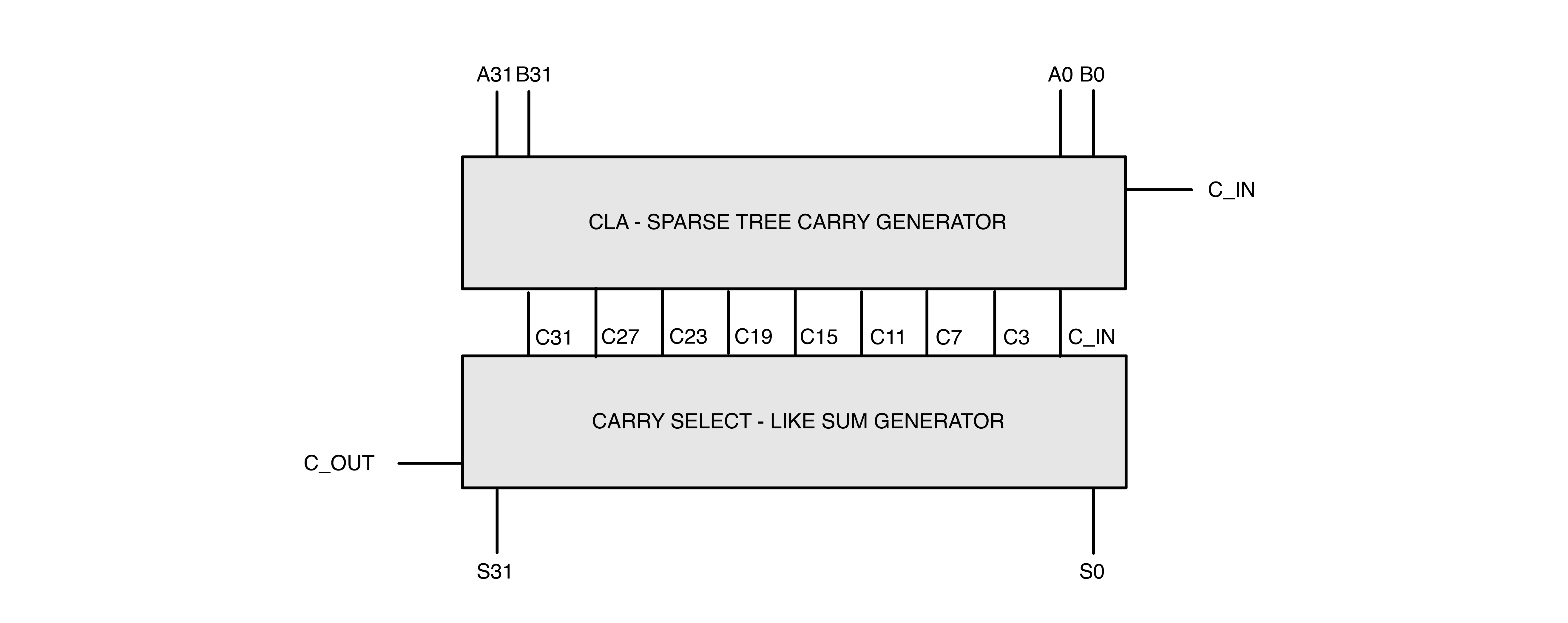
Comparison Unit



Shift Unit



Arithmetic Unit



Datapath

Features

- Forwarding
  - One stall case: load-use hazard
- Branch Handling
  - Branches assumed not taken; if taken, flush triggered (handled by Control Unit)

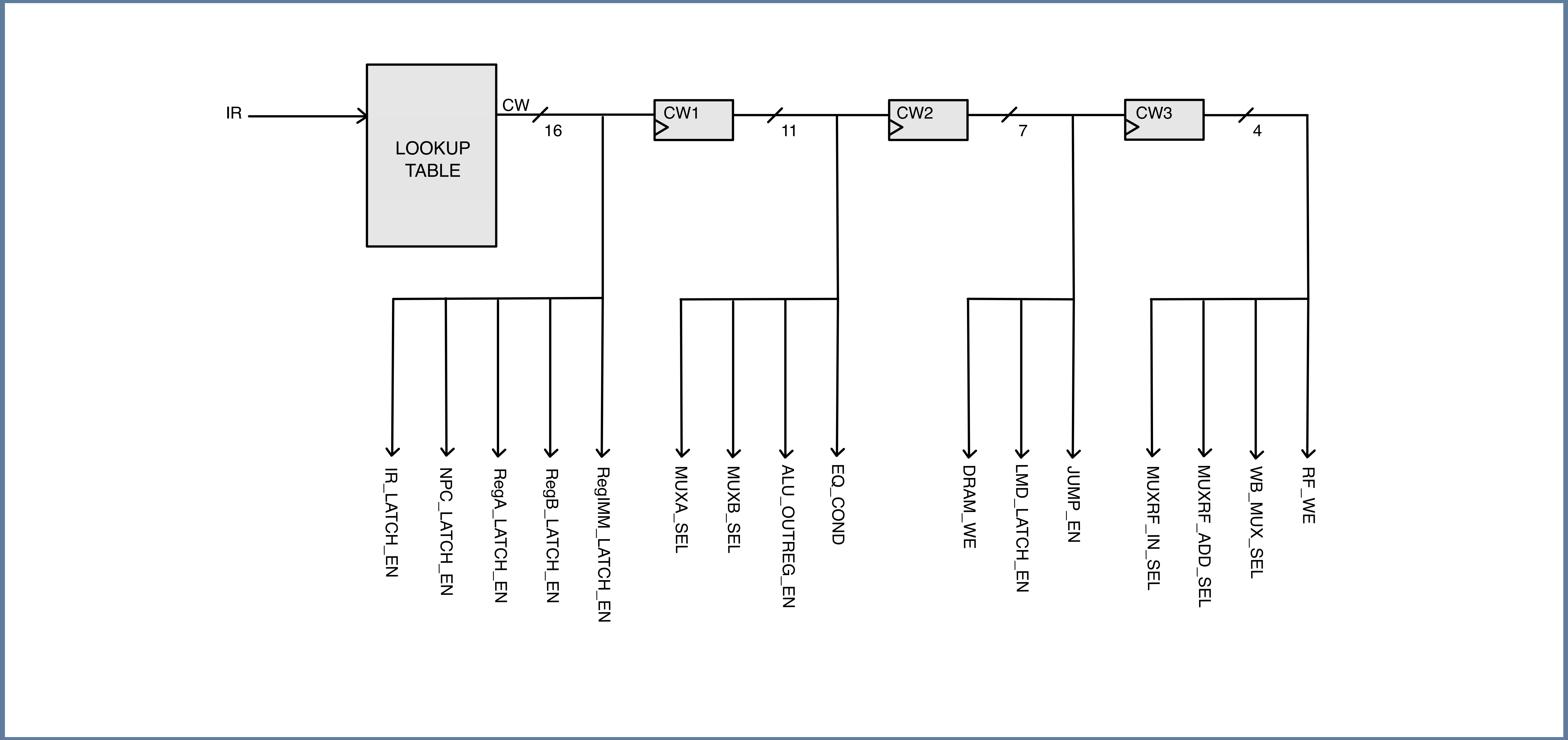
Control Unit

Overview

- Hardwired Control Unit
- Lookup table
    - 44 entries (26 instructions + NOPs)
- Control Word
    - 16 Control Signals
    - Forwarding of required control signals through stages

Control Unit

Block Diagram



Memory Subsystems

Overview

- Instruction Memory (IRAM)
    - 48 words of 32 bits
- Data Memory (DRAM)
    - 32 words of 32 bits
- Not synthesized

Simulation

Overview

- QuestaSim tool - testbench simulation
- Assembly -> Binary -> Memory conversion of instructions
- Scripts: compile.bash, sim.do, assembler.bash (dlxasm.pl, conv2memory)
- Test programs
    - Test 1: Loop Handling
    - Test 2: Complex Scenario
    - Test 3: Stalls and Forwarding

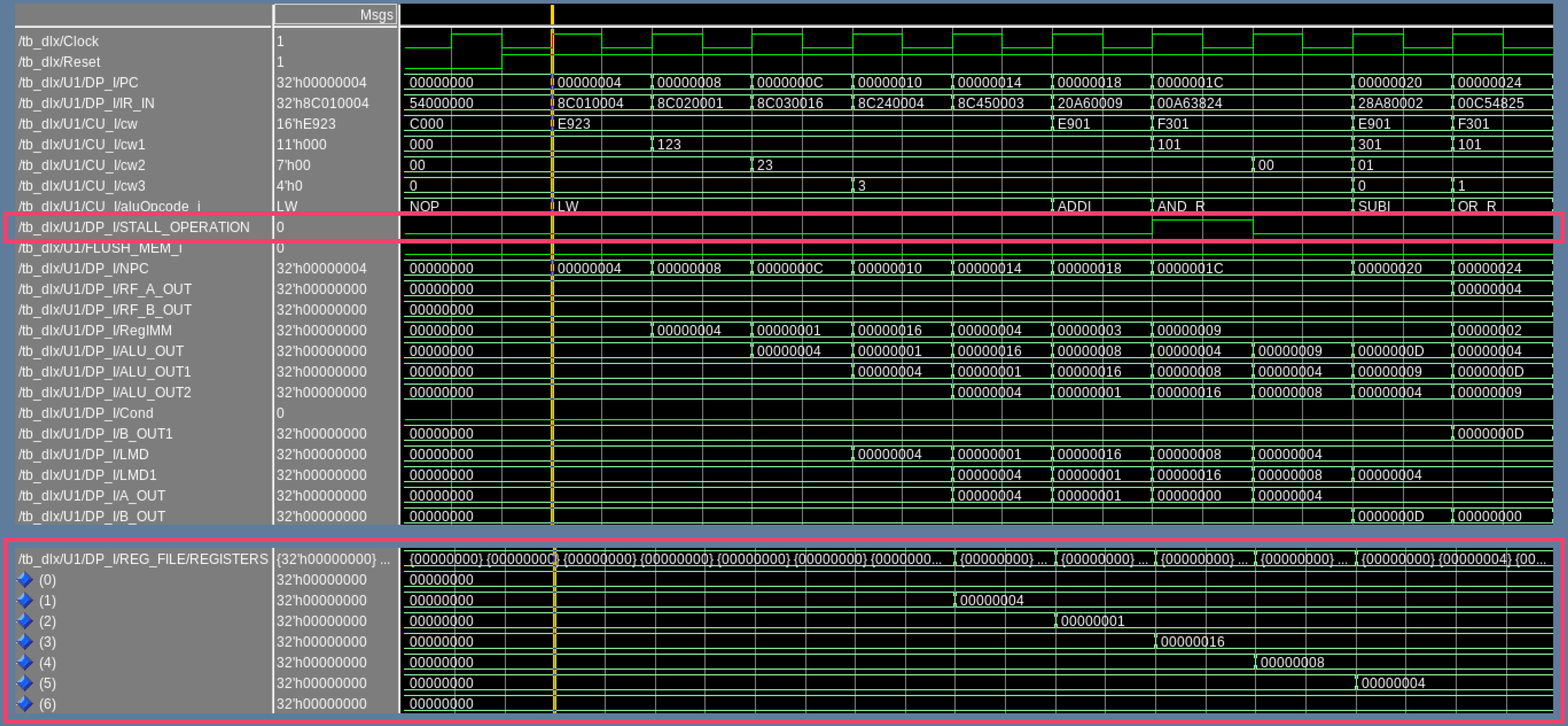
Simulation

Example (Test 3)

- lw r1, 4(r0)
- lw r2, 1(r0)
- lw r3, 22(r0)
- lw r4, 4(r1)
- lw r5, 3(r2)
- addi r6, r5, #9
- and r7, r5, r6
- subi r8, r5, #2
- or r9, r6, r5
- sw 1(r7), r7
- sw 1(r8), r8
- sw 1(r9), r9

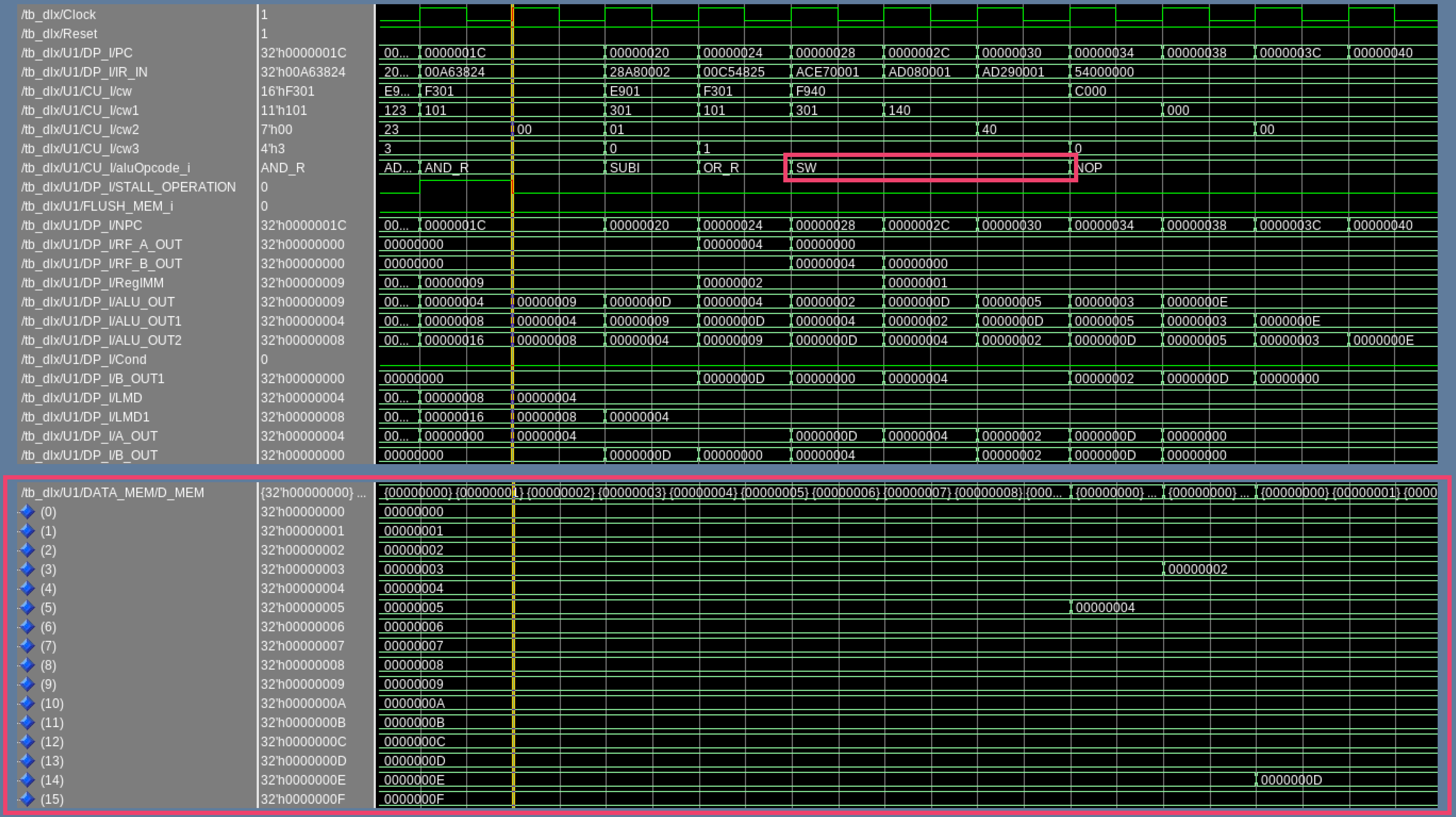
Simulation

Example - Waveforms



Simulation

Example - Waveforms



Synthesis

Overview

- Synopsys Design Compiler tool
- RTL to Gate-level
- **Timing**, Area, Power results for multiple clock constraints
- Datapath and Control Unit Synthesized separately
- Scripts: Bash, TCL, SDC

Synthesis

Results

Datapath

Clock Period	Frequency	Area	Slack	Power		
				Dynamic	Leakage	Total
0.18 ns	5.55 GHz	17731.83 μm^2	-1.24 ns (VIOLATED)	60.95 mW	355.71 μW	61.31 mW
0.19 ns	5.26 GHz	17715.33 μm^2	-1.22 ns (VIOLATED)	57.70 mW	355.58 μW	58.05 mW
1.50 ns	666.67 MHz	14279.94 μm^2	-0.10 ns (VIOLATED)	7.47 mW	297.32 μW	7.77 mW
1.60 ns	625.00 MHz	14186.58 μm^2	-0.01 ns (VIOLATED)	6.99 mW	295.92 μW	7.29 mW
1.75 ns	571.43 MHz	13594.99 μm^2	0.00 ns (MET)	6.33 mW	282.19 μW	6.61 mW
2.00 ns	500.00 MHz	13381.13 μm^2	0.00 ns (MET)	5.52 mW	277.10 μW	5.80 mW
2.50 ns	400.00 MHz	13318.09 μm^2	0.00 ns (MET)	4.41 mW	276.06 μW	4.69 mW
3.00 ns	333.33 MHz	13251.32 μm^2	0.01 ns (MET)	3.67 mW	274.56 μW	3.94 mW
3.125 ns	320.00 MHz	13253.98 μm^2	0.02 ns (MET)	3.51 mW	274.49 μW	3.79 mW
3.20 ns	312.50 MHz	13250.52 μm^2	0.06 ns (MET)	3.43 mW	274.31 μW	3.70 mW
3.25 ns	307.69 MHz	13251.06 μm^2	0.11 ns (MET)	3.38 mW	274.31 μW	3.65 mW
3.50 ns	285.71 MHz	13244.40 μm^2	0.37 ns (MET)	3.13 mW	274.20 μW	3.41 mW
4.00 ns	250.00 MHz	13243.87 μm^2	0.86 ns (MET)	2.74 mW	274.15 μW	3.02 mW
5.00 ns	200.00 MHz	13260.90 μm^2	1.79 ns (MET)	2.20 mW	273.44 μW	2.47 mW

Control Unit

Clock Period	Frequency	Area	Slack	Power		
				Dynamic	Leakage	Total
0.18 ns	5.55 GHz	106.67 μm^2	-0.01 ns (VIOLATED)	344.88 μW	2.28 μW	347.15 μW
0.19 ns	5.26 GHz	106.40 μm^2	0.00 ns (MET)	297.09 μW	2.23 μW	299.32 μW
1.50 ns	666.67 MHz	100.01 μm^2	1.30 ns (MET)	37.02 μW	2.09 μW	39.10 μW
1.60 ns	625.00 MHz	100.02 μm^2	1.40 ns (MET)	34.70 μW	2.08 μW	36.79 μW
1.75 ns	571.43 MHz	100.02 μm^2	1.55 ns (MET)	31.73 μW	2.09 μW	33.81 μW
2.00 ns	500.00 MHz	100.02 μm^2	1.80 ns (MET)	27.76 μW	2.09 μW	29.85 μW
2.50 ns	400.00 MHz	100.02 μm^2	2.30 ns (MET)	22.21 μW	2.09 μW	24.30 μW
3.00 ns	333.33 MHz	100.02 μm^2	2.80 ns (MET)	18.51 μW	2.09 μW	20.59 μW
3.125 ns	320.00 MHz	100.02 μm^2	2.93 ns (MET)	17.77 μW	2.09 μW	19.85 μW
3.20 ns	312.50 MHz	100.02 μm^2	3.00 ns (MET)	17.35 μW	2.09 μW	19.44 μW
3.25 ns	307.69 MHz	100.02 μm^2	3.05 ns (MET)	17.08 μW	2.09 μW	19.17 μW
3.50 ns	285.71 MHz	100.02 μm^2	3.30 ns (MET)	15.86 μW	2.09 μW	17.95 μW
4.00 ns	250.00 MHz	100.02 μm^2	3.80 ns (MET)	13.88 μW	2.09 μW	15.97 μW
5.00 ns	200.00 MHz	100.02 μm^2	4.80 ns (MET)	11.11 μW	2.09 μW	13.19 μW

Physical Design

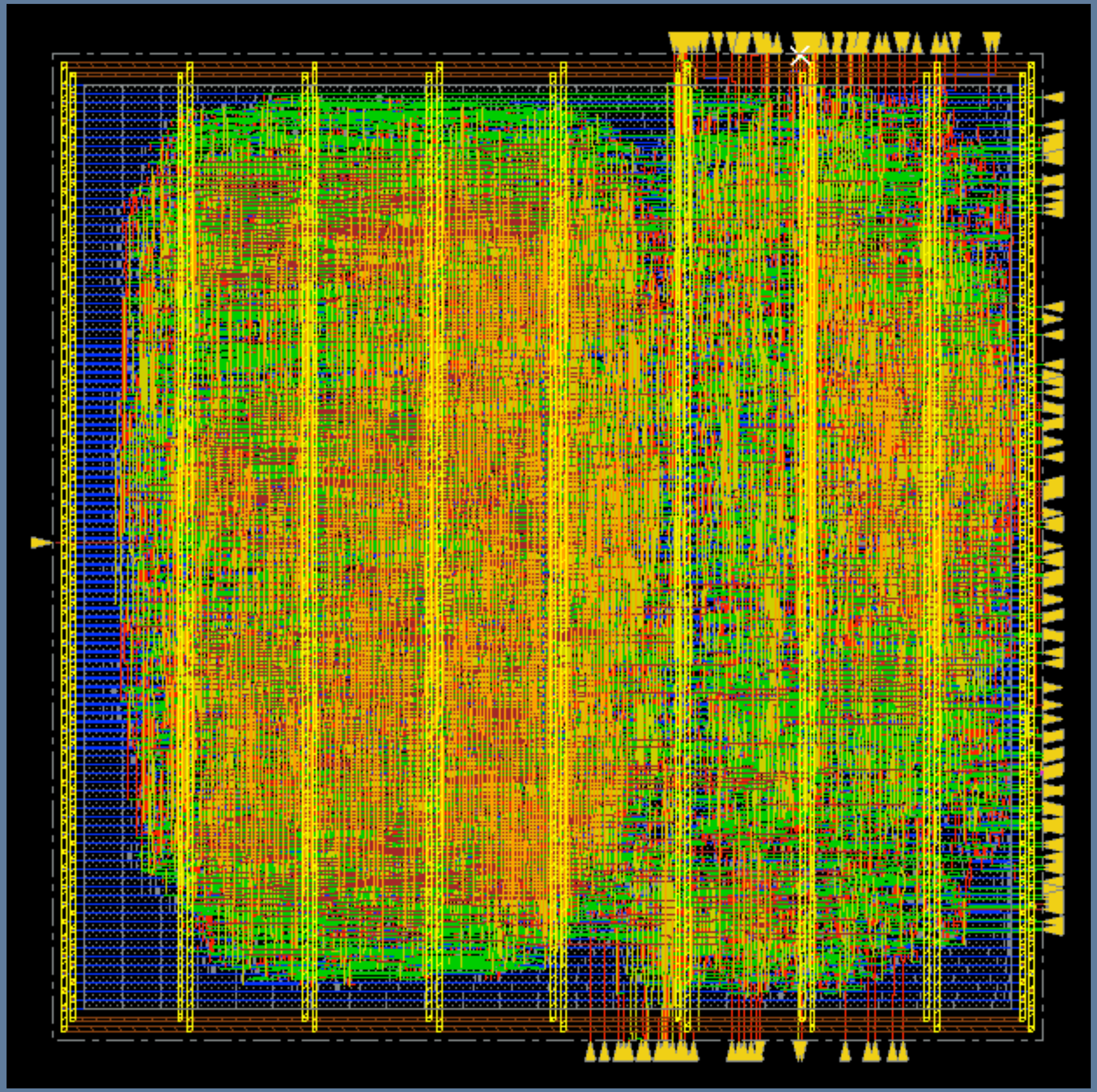
Overview

- Cadence Innovus tool
- Design Flow
  - Configuration & Design Import
  - Floorplanning & Power planning
  - Placement
  - Pre-CTS optimization, CTS, Post-CTS optimization
  - Routing & Finalization (Post-Routing optimization & Delay, Area, Power Analysis)

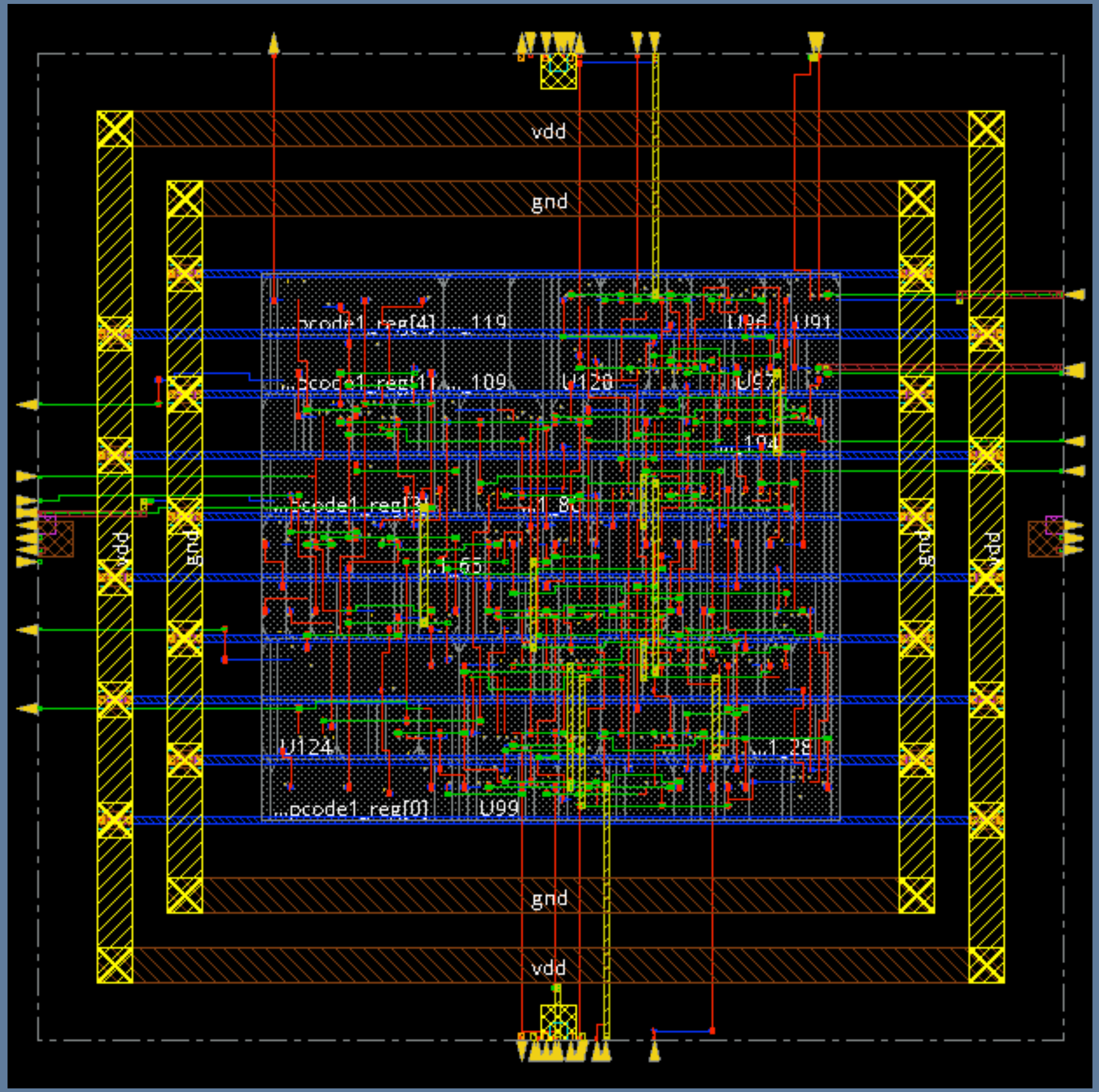
Physical Design

Results - Schematic

Datapath



Control Unit



Physical Design

Results - Analysis

- Control Unit
  - Area: 108.5 μm²
  - Power: 0.027 mW (63% combinational, 37% sequential)
  - Timing: met with setup WNS = 2.832 ns and hold WNS = 0.006 ns
- Datapath
  - Area: 13,429 μm²
  - Power: 5.95 mW (54% sequential, 32.5% combinational, 14% clock logic)
  - Timing: met with setup WNS = 0.917 ns and hold WNS = 0.007 ns

Physical Design

Results - Analysis

- Slight area and power increase due to fillers, routing, and parasitics wrt Synthesis stage
- No timing violations observed; all constraints met post-route optimization

Further Possible Implementations

- Extended Instruction Set
  - Floating-point operations
  - Subroutines
- Windowed Register File
- Out-of-Order Execution
- Branch prediction (BTB, BHT)

Thanks for listening!